Airline Data Analysis using Sql & python

1. Business problem

The airline runs flights across a large network of routes and serves thousands of passengers every day. For years, the focus has been on running reliable, comfortable flights — and that part is working. The real pressure right now is on profitability, and it is coming from multiple directions at once.

Costs are going up. Some routes are not planned well. Passenger demand keeps shifting in ways that are hard to predict. Any one of these would be manageable on its own. Together, they are squeezing the bottom line.
But when I looked at the data, one issue kept standing out above the rest — flights taking off with too many empty seats. That is not just an efficiency problem. Every empty seat is revenue the airline will never get back. The flight still costs the same to operate whether it is 30% full or 90% full. That gap is pure loss.

What makes it more frustrating is that the problem is not uniform. While some flights are flying nearly empty, other routes are completely packed. Passengers want to travel on certain corridors, but the airline is not allocating capacity or pricing tickets in a way that responds to that demand.

So the situation is this — underutilized capacity on routes that are not working, and missed revenue opportunity on routes that are.

2. Objectives

Before jumping into the data, I wanted to be clear about what I was actually trying to solve. Here are the four objectives I focused on:

i. Increase Occupancy Rate

The starting point was simple — too many flights are leaving with empty seats. I wanted to dig into booking patterns at the individual flight level to understand which flights are consistently underperforming and why. Because at the end of the day, every seat that flies empty is a fixed cost with zero return. Fixing occupancy is the most direct path to improving revenue per flight.

ii. Optimize Revenue and Pricing Strategy

This one went beyond just counting passengers. I wanted to understand how revenue is actually being generated — which routes are earning the most, which

ticket classes are contributing, and whether the current pricing is reflecting real demand or just sitting at a flat rate regardless of how full the flight is. The end goal here was to build a case for smarter pricing — including dynamic pricing that adjusts based on demand rather than a one-size-fits-all approach.

iii. Identify High-Demand Routes

Not all routes are created equal. Some corridors have passengers consistently booking them out while others are barely filling seats. I wanted to identify the routes where demand is strong and reliable — because those are the routes that deserve more investment, whether that means running more flights, deploying larger aircraft, or both. The data should be driving those capacity decisions, not assumptions.

iv. Improve Operational Efficiency

Running a half-empty flight costs almost the same as running a full one. That is the core inefficiency this objective targets. By identifying low-performing flights and routes with poor occupancy, the business can start making smarter calls — adjusting schedules, right-sizing aircraft, and cutting costs on routes that simply are not pulling their weight.

Analyzing occupancy rate

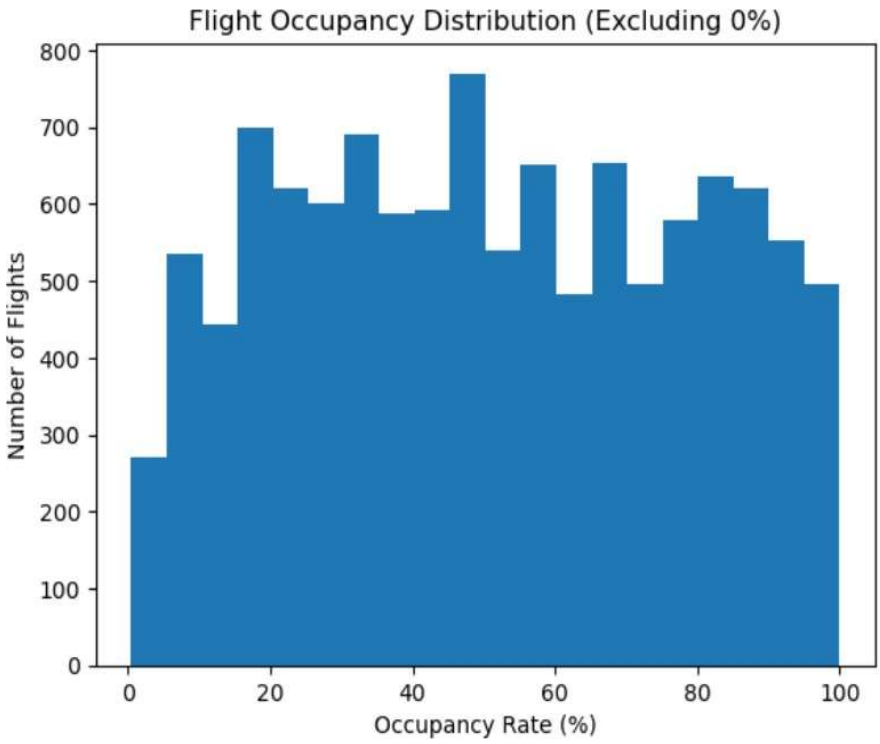

Figure 1 — Flight Occupancy Distribution
The first thing I wanted to understand was the overall shape of occupancy across the network — not just averages, but how spread out the numbers actually are.

What the chart shows is that occupancy is all over the place. There is no tight cluster around a healthy number. Flights are scattered across the full range — from barely any passengers to completely full — and that spread itself tells a story.

The part that stands out most is how many flights are sitting in the low to mid occupancy range. These are not grounded flights or special cases. These are regular scheduled services that are consistently not filling up. The aircraft is in the air, the crew is working, fuel is burning — but a significant chunk of seats are going empty on every single trip.

At the other end, some flights are running at high occupancy, which confirms that strong demand does exist on certain routes. Passengers are showing up — just not everywhere.

What this distribution is really showing is an imbalance. The demand is there in pockets, but it is not being matched with the right capacity or pricing across the board. That gap is where the inefficiency lives — and it is also where the biggest opportunity to improve profitability sits.

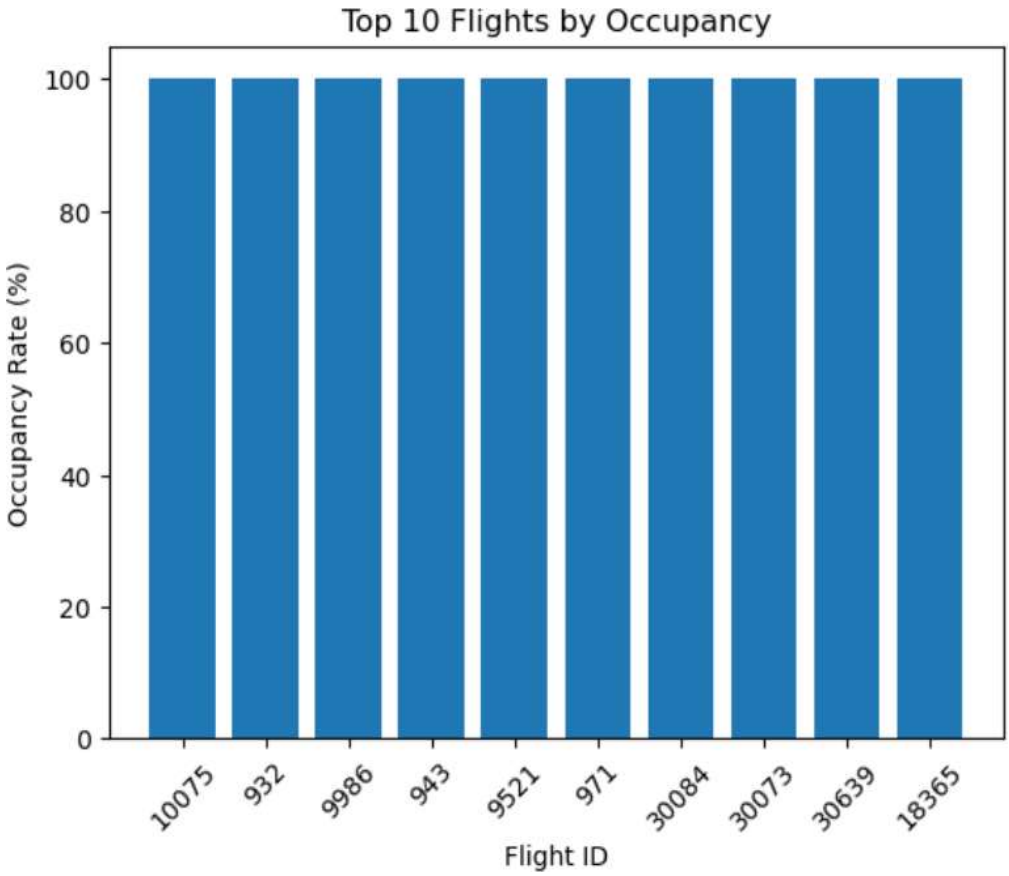

Figure 2 — Top 10 Flights by Occupancy
These 10 flights are doing everything right — and the chart makes that immediately obvious.

Every single one of them is operating at or right at 100% occupancy. Not close to full. Not usually full. Consistently, completely full. Passengers are booking these routes out and the airline is delivering on that demand every time.

What this tells me is that these are not just good flights — they are proof that the product works when demand and capacity are properly aligned. These routes have found their audience and that audience keeps coming back.

From a business standpoint, this is where the real opportunity is. When a flight is already running at 100% occupancy, the only way to earn more from that demand is to give it more room to grow. That means running these routes more frequently, switching to larger aircraft to carry more passengers per trip, or applying dynamic pricing that reflects just how popular these routes actually are. Right now the airline is likely leaving money on the table by capping what these routes can earn simply because there are not enough seats to sell.

These flights should not just be maintained — they should be the template the rest of the network is measured against.

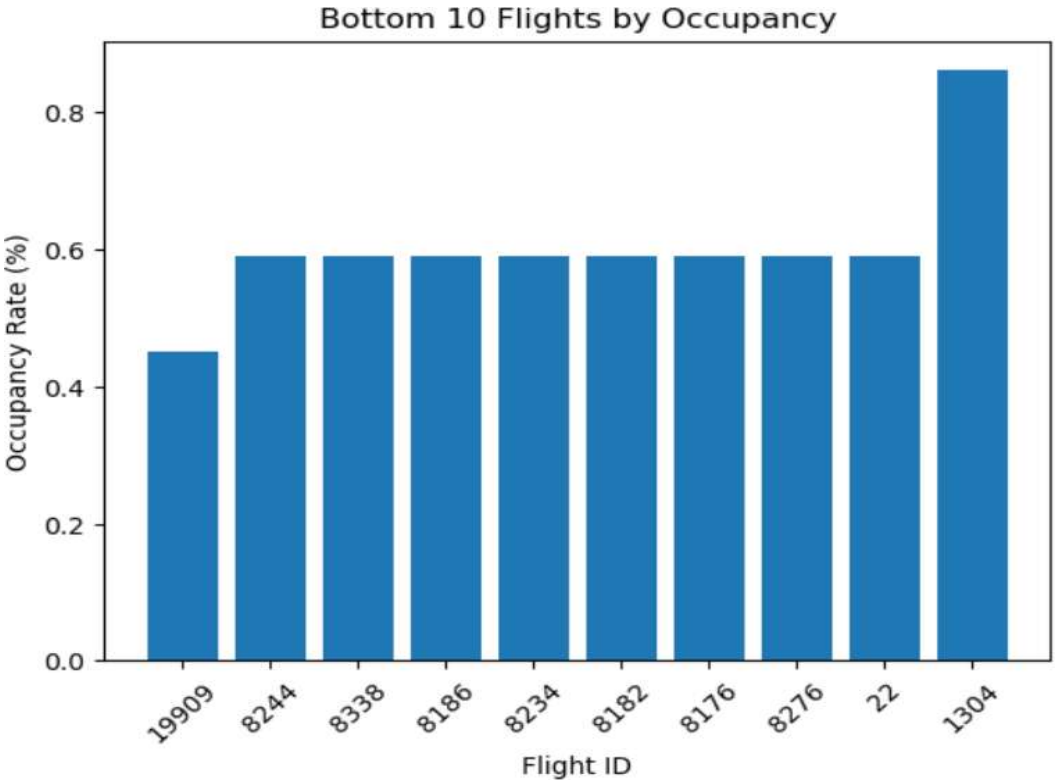

Figure 3 — Bottom 10 Flights by Occupancy
If the previous chart showed what good looks like, this one shows the other extreme — and it is hard to ignore.

Every flight in this group has an occupancy rate below 1%. That means for every 100 seats on the aircraft, fewer than one is filled. These flights are not struggling — they are essentially operating empty.

And the thing is, they are still flying. The fuel is still being burned, the crew is still being paid, the aircraft is still being maintained. All the costs of running a full flight are there — but almost none of the revenue is.

Technically these flights are not completely empty, but that distinction barely matters at this level. Whatever revenue they are generating is nowhere near enough to justify what it costs to keep them in the air. If anything, that makes it slightly more frustrating — because it means demand exists somewhere, it is just so thin that it is not making a real difference to the numbers.

These routes cannot stay the way they are. The business needs to make a call on each one:

- Cut how often they run and see if consolidating passengers onto fewer flights improves the economics
- Switch to smaller aircraft so the cost of operating the route actually matches the level of demand
- Test promotional pricing to find out whether lower fares can attract enough passengers to make these routes viable

If none of those moves work, the harder conversation about whether these routes should exist at all needs to happen. Keeping a near-empty flight on the schedule out of habit is not a strategy — it is just a cost.

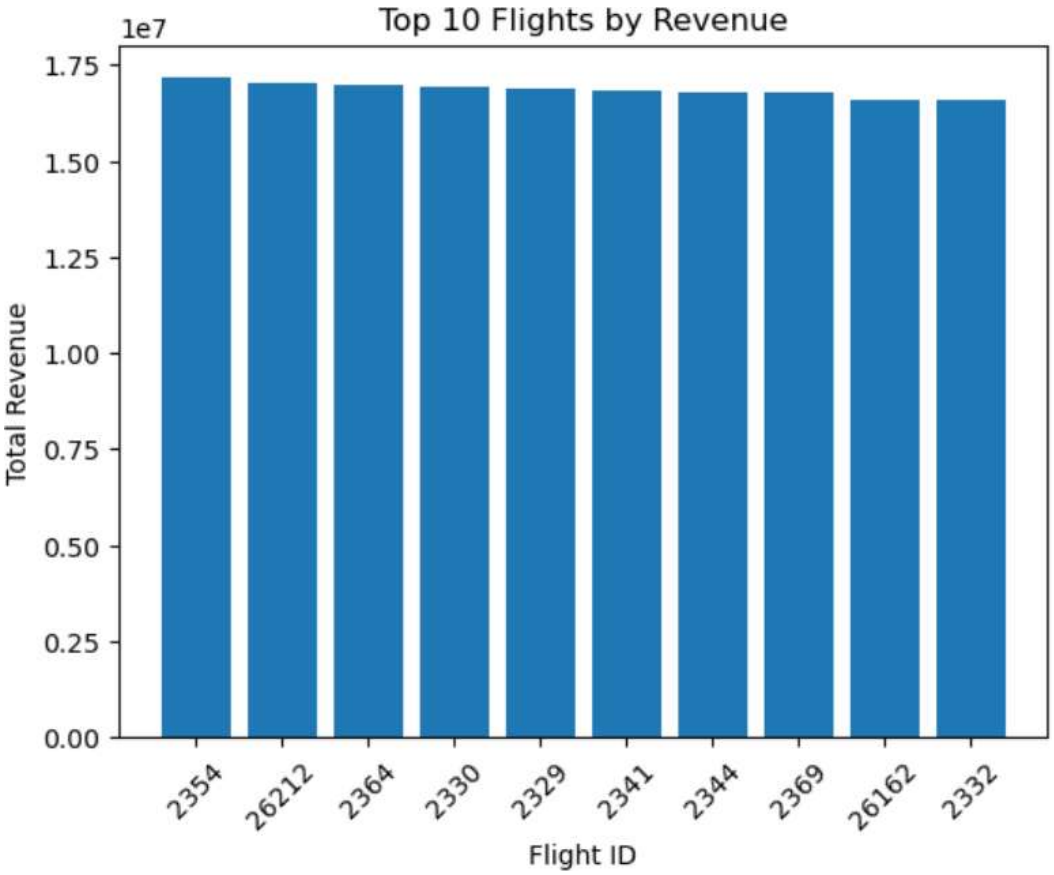

Figure 4 — Top 10 Flights by Revenue

This chart shifts the focus from how full the flights are to how much money they are actually generating — and the picture it paints is interesting.

The top 10 revenue-generating flights are all performing at a remarkably similar level. There is no single runaway outlier sitting far above the rest. They are all clustered together at the top, which tells you something important — this is not one lucky route carrying the entire business. These flights are consistently strong across the board.

That consistency is actually a good sign. It suggests that strong demand and sensible pricing are working together on these routes, not just one or the other. A flight can be full but still earn poorly if the tickets are priced too low. The fact that these flights are generating similar, high revenue levels means the combination is working.

These are the flights the business should be protecting above everything else. They are not just performing well right now — they are the foundation the rest of the network depends on. Any disruption to these routes — whether that is capacity cuts, pricing mistakes, or service quality dropping — would be felt immediately across the overall numbers.

The growth opportunity here is real. More frequency on these routes means more revenue from demand that is already proven. Smarter pricing means capturing more value from passengers who are clearly willing to pay. Better

service means protecting the loyalty that is keeping these routes consistently full.

These flights earned their spot at the top. The job now is to make sure they stay there and have room to grow.

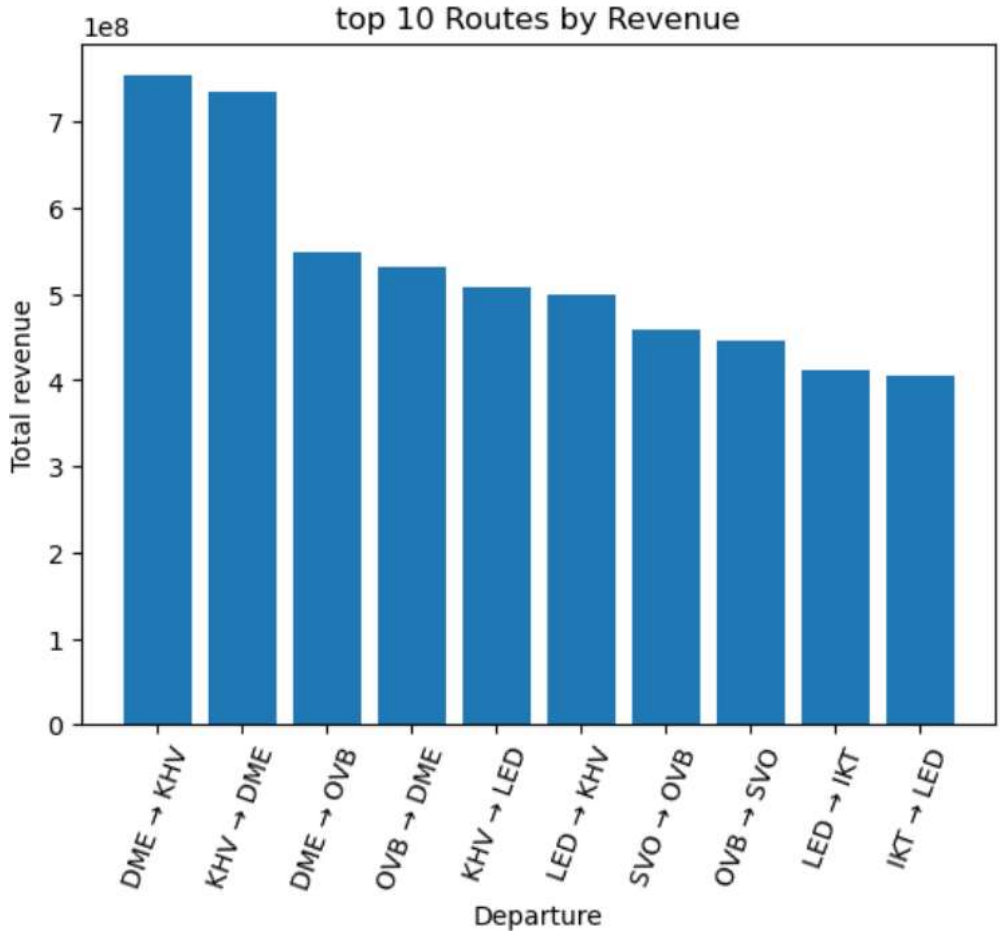

Figure 5 — Top 10 Routes by Revenue

This chart moves from individual flights to the bigger picture — which city-to-city corridors are actually driving the business.

Two routes immediately jump out at the top: **DME → KHV** and **KHV → DME**. These are the same corridor in both directions, and the fact that both versions appear at the very top of the revenue chart is not a coincidence. It tells you this is a genuinely high-demand route — passengers are travelling heavily in both directions, which is exactly what you want to see in a profitable corridor.

Below those two, the remaining routes show a gradual step-down in revenue rather than a sharp drop. That is actually a healthy pattern. It means the business is not dangerously dependent on just one or two routes — there is a solid group of consistently strong performers holding up the top of the network

together.

What these routes have in common is that demand and pricing appear to be working in the same direction. Passengers want to travel these corridors and the airline is capturing reasonable value from that demand. That combination is harder to build than it looks, which is why these routes deserve to be treated as assets worth investing in.

The expansion case here is straightforward. Routes that are already proven — strong passenger demand, consistent revenue, working pricing — are the lowest-risk place to put more resources. More flights on these corridors means more revenue from demand that does not need to be created from scratch.

Larger aircraft means more capacity for passengers who are already trying to book. Smarter pricing means capturing more value at peak times without losing the base demand that makes these routes reliable.
These are the routes the network should be built around.

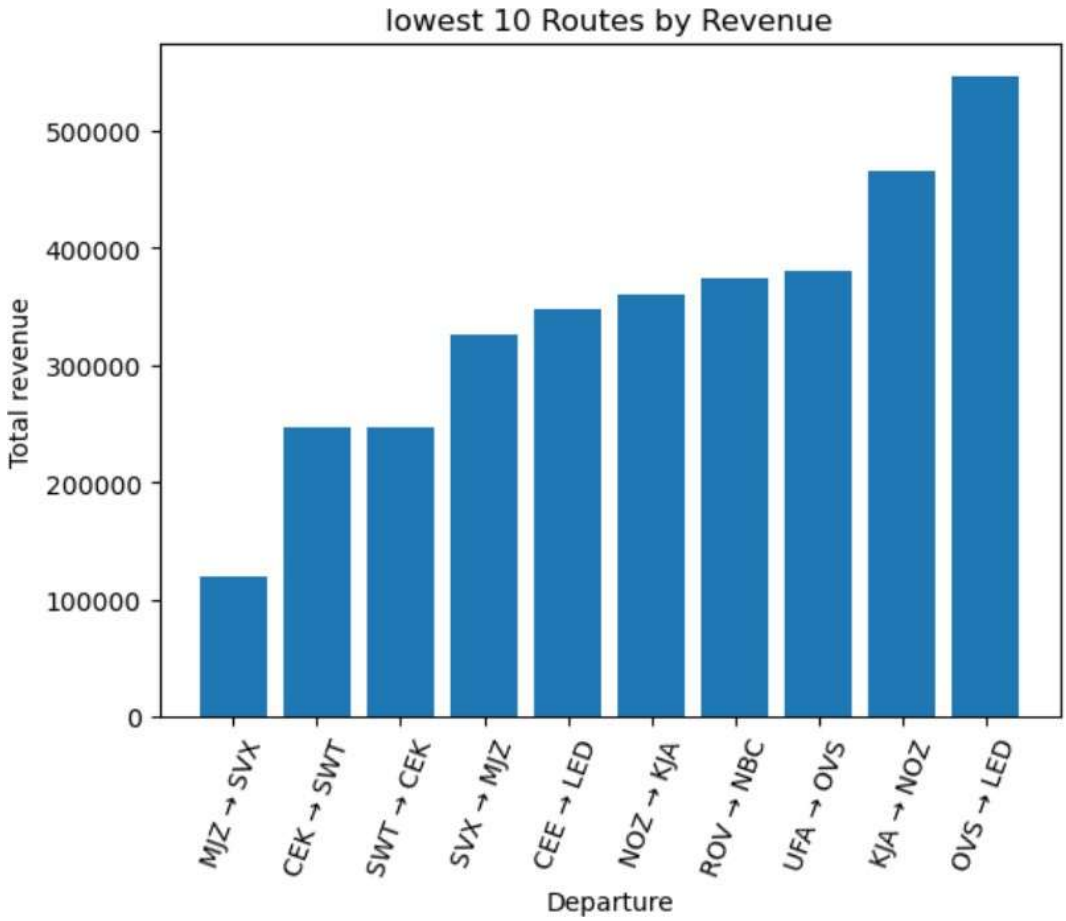

Figure 6 — Lowest 10 Routes by Revenue
After looking at where the money is coming from, it is equally important to look at where it is not — and this chart makes that picture very clear.

The gap between these routes and the top performers is not marginal. It is substantial. These routes are generating a fraction of what the strongest corridors earn, and that difference is not just a number on a chart — it represents real resources being deployed on routes that are not delivering meaningful returns.

What makes this more complicated is that the variation within this group suggests different underlying problems. Some of these routes are consistently weak across the board — low demand, low revenue, no real signs of improvement. Others are underperforming but sitting closer to the cutoff, which means the problem might be fixable with the right changes to pricing or scheduling rather than being a fundamental demand issue.

Either way, the common thread is that these routes are consuming capacity, crew time, and operational budget without contributing enough back to justify it. That is a quiet but very real drag on overall profitability — the kind that does not show up dramatically in any single report but adds up significantly over time.

The business needs to approach each of these routes with a clear-eyed question: is this route worth saving, and if so, what would actually fix it? For some, the answer might be reducing frequency to cut costs while keeping the route alive at a smaller scale. For others, a pricing adjustment tested over a defined period might be enough to move the needle. But for the ones that have consistently underperformed with no clear reason to expect that to change — the honest answer might be that continuing to operate them is a choice to keep losing money, and that choice deserves to be made deliberately rather than by default.

Business Recommendations

Based on everything the data showed, here is what I think the business should actually do differently.

Fix the low-occupancy flights — and stop being gentle about it. Flights running below meaningful occupancy levels are not a pricing problem or a marketing problem. They are a structural problem. Reducing how often they run, switching to smaller aircraft that actually match the level of demand, or consolidating passengers onto fewer departures are all moves that should have happened already. Every day these flights run the way they are is a day the business is paying full operating costs for a fraction of the return.

Give the high-demand routes room to grow. The flights running at or near 100% occupancy are the best signal in the entire dataset — passengers want these routes and they are booking them out. The response to that should not be to just maintain the status quo. Add more flights. Put larger aircraft on these corridors. The demand is already there and proven. All the business needs to do is give it more capacity to fill.

Stop using flat pricing across the board. Dynamic pricing is not a complicated idea — it just means charging more when demand is high and adjusting fares earlier when flights are struggling to fill. Right now the pricing does not appear to be responding to what the data is clearly showing about demand patterns. That needs to change. A flight that is going to be full regardless should be earning more per seat. A flight that is struggling to fill should be testing lower fares well before departure, not as a last-minute panic.

Treat the top revenue routes as the backbone of the business. DME → KHV and the other strong revenue corridors are not just performing well — they are holding up the overall numbers. These routes deserve targeted investment, whether that is more frequency, better service, or dedicated marketing to the passenger base that is already travelling them. Protecting what is working is just as important as fixing what is not.

Make a real decision on the weakest routes. Re-evaluating is not enough. Some of these routes need to be restructured significantly — different schedules, different aircraft, different pricing. Others may need to be discontinued altogether. The data makes a clear case that continuing to operate near-empty flights out of habit is costing the business real money. That conversation needs to happen with actual numbers on the table, not assumptions about future demand that has not materialized.

Project Summary

This project started with a straightforward question — where is this airline losing money and where is it leaving money on the table?

After going through the occupancy and revenue data, the answer became clear. The network is running two very different realities at the same time. On one end, flights are taking off nearly empty on a regular basis, burning through operational costs with almost nothing coming back in revenue. On the other end, certain flights and routes are completely packed, consistently generating strong revenue, and proving that real demand exists in the market.

The problem is not that the airline is doing everything wrong. The problem is that it is treating a very uneven network as if it were uniform — same approach to scheduling, same pricing logic, same capacity decisions regardless of what the demand data is actually showing.

The most important thing I took away from this analysis is that the gap between the best-performing flights and the worst-performing ones is not small. It is significant. And that gap represents a real, quantifiable cost that the business is absorbing every single day without a clear plan to address it.
The path forward is not complicated in theory — focus resources on routes where demand is proven, stop subsidizing routes that are not working, and build a pricing system that actually responds to how passengers are booking. None of that requires guesswork. The data is already there pointing in the right direction.

What this project really showed me is that data analysis in a business like this is not about producing charts. It is about giving decision-makers a clear, honest picture of what is working and what is not — so that the next call made on routes, capacity, or pricing is based on evidence rather than habit.